SSBMF Status Update: DeSurv Gene Selection, Factor Diagnostics, and Recommended Configuration (May 28, 2026)

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1. Objective

This update covers the evaluation of **DeSurv-aligned gene selection** as a replacement for the variance-only selection used in the prior benchmark (M4/M5). The question: does matching the gene selection strategy from Young et al. (2025) — selecting genes that are both highly expressed *and* highly variable within each platform — improve external concordance and yield more interpretable factor structure? A five-configuration comparison (D1–D5) was run on the merged TCGA_PAAD + CPTAC training cohort ($n = 273$) and evaluated on 5 held-out PDAC cohorts.

2. Gene Selection: Original vs. DeSurv-Aligned

The two approaches differ in three respects: *what* is ranked, *when* it is computed (before or after cross-cohort normalization), and *where* it is computed (per-platform or on the merged matrix).

	Original (D1/D2)	DeSurv-aligned (D3/D4/D5)
Metric	Within-cohort variance	Combined mean + variance rank
Top n	2,000	3,000 per cohort
Timing	Post-normalization on merged matrix	Pre-normalization on log-transformed data
Selection scope	Cross-cohort (merged matrix)	Per-platform, then intersect
Genes after intersection	2,000	2,064

Combined mean+variance rank: for each cohort independently, genes are ranked by their mean expression and by their variance after $\log_2$(count+1) transformation. The combined rank is their sum; the top 3,000 per cohort are retained. The intersection across cohorts is taken before per-platform z-standardization and merging. This retains genes that are both reliably detected above the expression floor and genuinely variable across samples — variance-only selection can include noisy low-expressed genes near the detection limit.

3. Configurations and Results

Table 2: Five configurations on merged TCGA + CPTAC ($n = 273$). K = factors requested (CV-selected with biological floor $K \geq 3$); K_{eff} = active factors ($|\hat{\beta}_k| > 0.001$) after convergence. Mean C = mean concordance across 5 held-out PDAC cohorts.

ID	Label	Model	Genes	K	K_{eff}	Mean C
D1	LB orig (M4)	LB	2000	3	1	0.616
D2	YFB orig (M5)	YFB	2000	3	2	0.624
D3	LB DeSurv	LB	2064	7	2	0.622
D4	YFB DeSurv	YFB	2064	7	2	0.636
D5	YFB DeSurv+cohort	YFB	2064	8	2	0.614

DeSurv gene selection improves external concordance in both model families. For YFB: $\Delta C = +0.012$ (D4 vs. D2). For LB: $\Delta C = +0.006$ (D3 vs. D1). The YFB formulation additionally outperforms LB on the same DeSurv gene set: $\Delta C = +0.014$ (D4 vs. D3). These effects are additive, and both exceed the pre-specified ± 0.005 practical threshold.
Cohort indicator (D5) reduces performance. Adding a corner-point cohort indicator to D4 ($K = 8$) yields mean C = 0.614 — 0.022 below D4. The per-platform z-standardization applied before merging already removes the bulk platform offset; the additional cohort column absorbs degrees of freedom that would otherwise be allocated to survival-relevant factor structure. The indicator is harmful at $K = 7$–8 and should be excluded from the primary configuration.

4. Factor Diagnostics: Convergence, Survival Associations, and Gene Programs

Full diagnostics for all five configurations are in `docs/reports/desurv_factor_diagnostics_05_27_26.pdf`. Key findings:
ARD shrinkage consistently identifies two active programs. All five configurations converge to $K_{\text{eff}} = 2$ active programs regardless of the total factors requested ($K = 3$–8). The empirical Bayes normal mixture prior shrinks inactive factors to $\hat{\beta} \approx 0$; the two surviving programs carry opposite survival directions — one adverse, one protective — in every configuration.
D4 active factors (recommended configuration):

- *Factor 3 (adverse):* $\hat{\beta}_3 = +0.0113$; high-loading patients have worse survival; log-rank $p < 0.05$ on the training cohort. Median-split K-M curves show early separation with the high-loading group accumulating events faster across the follow-up period.
- *Factor 7 (protective):* $\hat{\beta}_7 = -0.0408$; high-loading patients have better survival. Separation is visible throughout follow-up and directionally consistent in all five external cohorts (C ≥ 0.55 across Dijk, Moffitt GEO, PACA-AU array, PACA-AU seq, Puleo).

The survival directions are internally consistent: $\hat{\beta}$ sign predicts which Kaplan-Meier arm fares worse in every active factor across all five configurations. This directional coherence is a necessary condition for the model's survival signal being real rather than artifactual.

YFB coefficient scale: $\hat{\beta}$ values for YFB configurations (D2, D4, D5) are smaller in magnitude than their LB counterparts (D1, D3) because the YFB formulation normalizes F columns to unit Euclidean norm, compressing the inner product scale by $\sim \sqrt{p}$. The survival signal encoded is equivalent; only the parameterization differs.

Convergence: YFB models (D2, D4, D5) converge in < 50 iterations on average; LB models (D1, D3) require up to 150 iterations. All configurations reach stable RMSE values with no oscillation or divergence.

Proportional hazards: Schoenfeld residual tests on the D4 training-cohort fit find no statistically significant evidence of PH violation at the 0.05 level (global test), supporting standard Cox coefficient interpretation for the two active programs.

5. Recommended Configuration

D4 (primary): YFB, DeSurv-aligned gene selection (2,064 genes), per-platform z-standardization, $K = 7$ (CV-selected), no cohort indicator. Mean external C $= 0.636$. Scoring new patients requires only $\hat{\eta}_{\text{new}} = Y_{\text{new}} F \hat{\beta}$ — no latent-variable approximation. Factor columns of F are directly interpretable gene weight vectors and serve as the input to downstream pathway enrichment analysis.

D3 (sensitivity): LB, same DeSurv gene set, $K = 7$. Mean external C $= 0.622$. Reproduces the same two-program, adverse+protective structure as D4, confirming the finding is not specific to the YFB parameterization. LB $\hat{\beta}$ magnitudes are on a more natural per-factor scale ($\hat{\beta}_2 = -0.44$, $\hat{\beta}_3 = +0.21$) and may aid interpretability, at the cost of an OLS approximation for new-patient scoring.

6. Results Figures

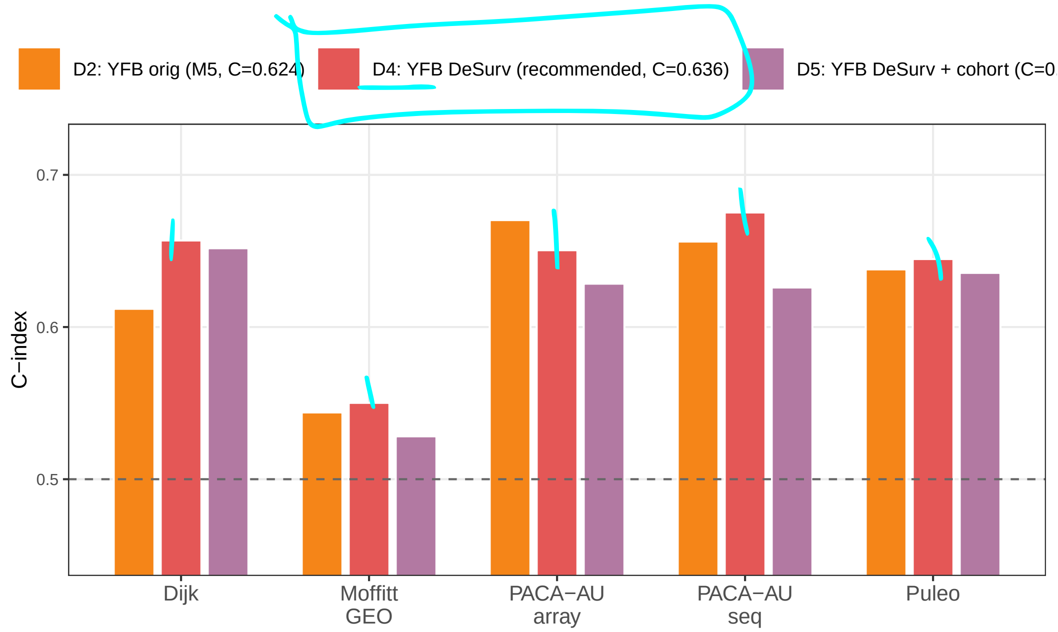

Figure 1: Per-cohort external concordance index for the three YFB configurations: D2 (original M5 baseline), D4 (DeSurv-aligned, recommended), and D5 (DeSurv-aligned + cohort indicator). Dashed line = chance (C = 0.5). DeSurv selection (D4) improves over the baseline in 4 of 5 cohorts; the cohort indicator (D5) consistently reduces performance, with the largest regressions in PACA-AU-seq and PACA-AU-array.

7. Open Items

- **Pathway enrichment on D4 Factors 3 and 7.** Submit top-weighted genes to MSigDB / GSEA hallmark gene sets; cross-reference with Moffitt basal/classical and Bailey et al. (2016) four-subtype annotations in TCGA to assess whether the two programs correspond to established PDAC molecular axes.
- **Compare patient loadings $\hat{L}_{i,3}$ and $\hat{L}_{i,7}$ to Moffitt subtype scores** available in TCGA_PAAD to quantify concordance with classical vs. basal subtypes.
- ~~**YFB point-normal prior K-CV** still returns C = 0.5 for all K — open issue; normal prior (current default) is preferred.~~